

“THE DISCUSSION OF
NATURAL PROBLEMS
OUGHT TO BEGIN... WITH
EXPERIMENTS AND
DEMONSTRATIONS.”

Galileo Galilei, Italian astronomer and physicist, from *The Authority of Scripture in Philosophical Controversies*

Galileo demonstrated his “law of falling bodies” by rolling a ball down an inclined plane and measuring its rate of acceleration.

De Magnete also claimed that **magnetism and electricity are two distinct kinds of force**. To show the **properties of static electricity**, Gilbert created a **versorium**—the first **electroscope**, comprising a freely rotating unmagnetized needle on a stand. The versorium’s needle was attracted to static-charged amber, as if it were a compass needle moved by magnetism. Gilbert incorrectly inferred that gravity was a magnetic force, and that Earth’s magnetism held the Moon in its orbit.

In the same year, Italian friar and astronomer **Giordano Bruno** (1548–1600) was **burned at the stake** by the Roman Inquisition on charges of heresy. It is possible that he was originally arrested purely for his unconventional theological beliefs, but it is more likely that his **scientific views** were the real reason for the Inquisition’s wrath. Bruno’s theory of the cosmos went a step further than Copernicus’s (see 1543), and was potentially more threatening to the Church’s authority: he believed that **the Sun**, far from being the center of the Universe, **was just a star like any other**, and that it was possible that **Earth was not the only world inhabited** by intelligent life.

ENGLISH ASTRONOMER AND MATHEMATICIAN THOMAS HARRIOT (1560–1621) was fascinated by the behavior of light. In 1602, he studied the relationship between the different angles produced as light is refracted, or bent, when passing from one medium to another, such as from air to water. Now known as the **law of refraction**, this phenomenon had first been discovered by **Persian mathematician Ibn Sahl** in 984. Unfortunately, Harriot did not publish his findings, and the principle is now known as Snell’s Law, after Willebrord Snellius (see 1621–24), who rediscovered the idea around 20 years later.

The following year, naturalist **Federico Cesi** (1585–1630) founded a scientific society called

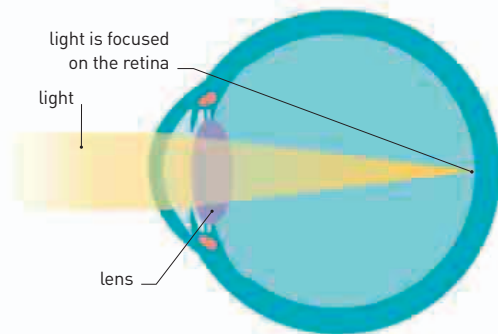
Accademia dei Lincei (Academy of the Lynx-eyed) in Rome—a successor to the Academia Secretorum Naturae (Academy of the Secrets of Nature), which had been founded in 1560 but forced to disband. The Accademia dei Lincei later became the national academy of Italy.

The story of **Galileo Galilei** (see 1611–13) dropping balls of different weights from the top of the Leaning Tower of Pisa, Italy, to ascertain the rate of their fall may or may not have been true; however, it is known that in 1604 he hypothesized for the first time that **bodies made of the same material and falling through the same medium would fall at the same speed**, regardless of their mass. This idea contradicted the prevalent Aristotelian theory

JOHANNES KEPLER (1571–1630)



Born in Germany, Johannes Kepler studied at the University of Tübingen, where he encountered the ideas of Copernicus. He worked as a teacher in Graz, Austria, before moving to Prague to study with Tycho Brahe in 1600. He remained there as Imperial Astronomer after Brahe’s death, until political and family problems forced him to leave 12 years later.



Human eye
The eye sees by allowing light in through a lens near the front. The lens projects the light onto the retina at the back, and focuses it to give a clear though inverted image.

that the heavier the object, the faster it would fall. Galileo published the final version of his **law of falling bodies** in 1638.

Best known as an astronomer, **Johannes Kepler** was also a pioneer in the field of optics, publishing *Astronomiae Pars Optica* (*The Optical Part of Astronomy*) in 1604. In addition to describing astronomical instruments, he devoted much of the text to optical theory, including explanations of **parallax** (the apparent change in position of a heavenly body when viewed from different points), **reflections in flat and curved mirrors**, and **the principle of the**

pinhole camera. He examined the **optics of the human eye**, describing how the lens reverses and inverts the image projected onto the retina, and suggested that this is corrected in the brain.

In 1604, Italian surgeon and anatomist **Hieronymus Fabricius** (see 1594–95) published the results of his dissections of various animals’ fetuses, establishing **embryology** as a new field of study. He showed various stages of fetal development, and combined these studies with his work on the circulation of blood to produce one of the first studies of **embryonic circulation**.

“DISCOVER THE FORCE
OF THE HEAVENS, O MEN:
ONCE RECOGNIZED,
IT CAN BE PUT TO USE.”

Johannes Kepler, German astronomer, from *De Fundamentis*, 1601

1600 Giordano Bruno is burned at the stake for his heretical views

1600 Johannes Kepler begins work as Tycho Brahe’s assistant in Prague

1602 Thomas Harriot arrives at the law of refraction

1603 The Accademia dei Lincei is founded in Rome

1604 Galileo Galilei begins to formulate his law of falling bodies

1604 Johannes Kepler describes how the eye focuses light in *Astronomiae Pars Optica*

1604 Hieronymus Fabricius begins his study of blood circulation in embryos